CORE PURE / COMPLEX NUMBERS

Complex Numbers Basics

Worked solutions 8 questions · 62 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Instructions

Cambridge International: this topic belongs to A Level Mathematics (9709), Paper 3 and is assumed knowledge for Further Mathematics 9231.

Questions

Question 1	*7 marks*	2
Question 2	*5 marks*	3
Question 3	*5 marks*	4
Question 4	*7 marks*	5
Question 5	*8 marks*	7
Question 6	*10 marks*	9
Question 7	*11 marks*	11
Question 8	*9 marks*	13

1. **(a)** Let $z = x + iy$, where $x, y \in \mathbb{R}$.

(i) Write down z^*, the complex conjugate of z, in terms of x and y. **[1]**

(ii) Hence show that the product of a complex number and its conjugate is always real. **[2]**

(b) The complex number w satisfies

$$2w + 12 = 5w^* + 14i$$

(i) Find w, giving your answer in the form $a + bi$, where $a, b \in \mathbb{R}$. **[3]**

(ii) Calculate $w^2(w^*)^2$. **[1]**

Solution

(a) (i) The complex conjugate is

$$z^* = x - iy$$

(ii) Using part (a)(i) and $i^2 = -1$:

$$\begin{aligned} zz^* &= (x + iy)(x - iy) \\ &= x^2 - ixy + ixy - i^2y^2 \\ &= x^2 + y^2 \in \mathbb{R} \end{aligned}$$

(b) (i) Write $w = x + iy$, where $x, y \in \mathbb{R}$. Then $w^* = x - iy$, so the given equation becomes

$$\begin{aligned} 2(x + iy) + 12 &= 5(x - iy) + 14i \\ 2x + 12 + 2iy &= 5x + (14 - 5y)i \end{aligned}$$

Equating real and imaginary parts gives

$$\begin{aligned} 2x + 12 = 5x \quad &\Longrightarrow \quad x = 4 \\ 2y = 14 - 5y \quad &\Longrightarrow \quad y = 2 \end{aligned}$$

Hence $w = 4 + 2i$, in the form $a + bi$ with $a = 4$, $b = 2$.

(ii) Since $w^* = 4 - 2i$:

$$\begin{aligned} w^2(w^*)^2 &= (4 + 2i)^2(4 - 2i)^2 \\ &= (12 + 16i)(12 - 16i) \\ &= 12^2 + 16^2 = 400 \end{aligned}$$

2. The complex number z satisfies

$$4(z+2i) = (1-2i)z^*$$

where z^* denotes the complex conjugate of z.
Determine z, giving your answer in the form $a+bi$, where a and b are real. **[5]**

Solution

Let $z = a+bi$, where $a, b \in \mathbb{R}$. Then $z^* = a-bi$, so the given equation becomes

$$4(a+bi+2i) = (1-2i)(a-bi)$$

$$4a + (4b+8)i = a - bi - 2ai + 2bi^2$$

$$4a + (4b+8)i = (a-2b) + (-2a-b)i$$

Equating real and imaginary parts gives

$$4a = a - 2b \qquad \Longrightarrow \qquad 3a + 2b = 0$$

$$4b + 8 = -2a - b \qquad \Longrightarrow \qquad 2a + 5b = -8$$

From the first equation, $a = -\frac{2b}{3}$. Substituting into the second:

$$2\left(-\frac{2b}{3}\right) + 5b = -8$$

$$\frac{11b}{3} = -8$$

$$b = -\frac{24}{11}$$

$$a = -\frac{2}{3}\left(-\frac{24}{11}\right) = \frac{16}{11}$$

So the answer in the form $a+bi$ is

$$z = \frac{16}{11} - \frac{24}{11}i$$

3. The complex numbers z_1 and z_2 are given by

$$z_1 = \sqrt{3}\left(\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right)$$

and

$$z_2 = 2\left(\cos\left(\frac{5\pi}{6}\right) - i\sin\left(\frac{5\pi}{6}\right)\right)$$

(a) State the exact value of

(i) $|z_1 z_2|$,

(ii) the principal argument of $z_1 z_2$, taking $-\pi < \arg(z_1 z_2) \leq \pi$.

[2]

Let $w = z_1 z_2$.

(b) Determine

(i) the smallest positive integer n for which w^n is a positive real number,

(ii) the exact value of $|w^n|$ for this value of n.

[3]

Solution

(a) (i) For a product, the modulus is the product of the moduli and the argument is the sum of the arguments, adjusted by a multiple of 2π to lie in the principal range:

$$|z_1 z_2| = |z_1||z_2|, \qquad \arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$

Here $|z_1| = \sqrt{3}$ and $|z_2| = 2$, so

$$|z_1 z_2| = \sqrt{3} \times 2 = 2\sqrt{3}$$

(ii) The minus sign before $i\sin\left(\frac{5\pi}{6}\right)$ means that z_2 has argument $-\frac{5\pi}{6}$. Adding the arguments gives a value in the required principal range:

$$\arg(z_1 z_2) = \frac{\pi}{6} - \frac{5\pi}{6} = -\frac{2\pi}{3}$$

(b) (i) De Moivre's theorem is

$$[r(\cos\theta + i\sin\theta)]^n = r^n(\cos n\theta + i\sin n\theta)$$

so raising w to the power n multiplies its argument by n. For w^n to be positive real, this angle must be a whole multiple of 2π, so

$$\frac{n\left(-\frac{2\pi}{3}\right)}{2\pi} = -\frac{n}{3} \in \mathbb{Z}$$

Thus n must be divisible by 3, and the smallest positive value is

$$n = 3$$

(ii) Using $|w| = 2\sqrt{3}$ from part (a)(i) and $n = 3$ from part (b)(i):

$$|w^3| = |w|^3 = (2\sqrt{3})^3 = 24\sqrt{3}$$

4. In this question the argument of a complex number is taken in the range $0 \leq \arg z < 2\pi$.
Complex numbers z_1 and z_2 are defined by $z_1 = 4 + 3i$ and $z_2 = -12 + 5i$.

(a) Determine $z_1 z_2$, giving your answer in the form $a + bi$. **[2]**

(b) Express z_2 in modulus-argument form. **[2]**

(c) By considering the formula for $\tan(A - B)$, verify that $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$. **[3]**

Solution

(a) Expanding and using $i^2 = -1$:

$$\begin{aligned} z_1 z_2 &= (4 + 3i)(-12 + 5i) \\ &= -48 + 20i - 36i + 15i^2 \\ &= -63 - 16i \end{aligned}$$

This is in the form $a + bi$ with $a = -63$, $b = -16$.

(b) The modulus is

$$|z_2| = \sqrt{(-12)^2 + 5^2} = 13$$

z_2 lies in the second quadrant, so its argument is

$$\theta = \pi - \arctan\left(\frac{5}{12}\right)$$

Thus the modulus-argument form is

$$z_2 = 13\left[\cos\left(\pi - \arctan\left(\frac{5}{12}\right)\right) + i\sin\left(\pi - \arctan\left(\frac{5}{12}\right)\right)\right]$$

(c) Let $\alpha = \arctan(\frac{3}{4})$ and $\beta = \arctan(\frac{5}{12})$. z_1 lies in the first quadrant, z_2 in the second and $z_1 z_2 = -63 - 16i$ in the third, so

$$\arg(z_1) = \alpha$$

$$\arg(z_2) = \pi - \beta$$

$$\arg(z_1 z_2) = \pi + \arctan\left(\frac{16}{63}\right)$$

The compound angle formula is

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B} \qquad \left(A \pm B \neq \left(k + \tfrac{1}{2}\right)\pi\right)$$

With $\tan\alpha = \frac{3}{4}$ and $\tan\beta = \frac{5}{12}$, this gives

$$\tan(\alpha - \beta) = \frac{\frac{3}{4} - \frac{5}{12}}{1 + \frac{3}{4} \times \frac{5}{12}} = \frac{16}{63}$$

Since $0 < \alpha - \beta < \frac{\pi}{2}$, we have $\alpha - \beta = \arctan(\frac{16}{63})$. Therefore

$$\begin{aligned}\arg(z_1) + \arg(z_2) &= \alpha + \pi - \beta \\ &= \pi + \arctan\left(\frac{16}{63}\right) \\ &= \arg(z_1 z_2)\end{aligned}$$

5. The complex numbers z_1 and z_2 satisfy

$$z_1 = -3 + 2i$$
$$|z_1 z_2| = 26\sqrt{2}$$
$$\arg(z_1 z_2) = -\frac{\pi}{4}$$

(a) Express z_1 in the form $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $-\pi < \theta \leq \pi$.
Give r exactly and θ in radians to 4 significant figures. **[2]**

(b) Determine z_2, giving your answer in the form $a + bi$, where a and b are integers. **[6]**

Solution

(a) z_1 lies in the second quadrant, as its real part is negative and its imaginary part is positive, so

$$r = |z_1| = \sqrt{(-3)^2 + 2^2} = \sqrt{13}$$
$$\theta = \pi - \arctan\left(\frac{2}{3}\right)$$

The required form is

$$z_1 = \sqrt{13}(\cos 2.554 + i\sin 2.554) \qquad (4 \text{ s.f.})$$

(b) Write $z_2 = a + bi$, where a, b are real. Multiplying gives

$$\begin{aligned} z_1 z_2 &= (-3 + 2i)(a + bi) \\ &= -3a - 3bi + 2ai + 2bi^2 \\ &= (-3a - 2b) + (2a - 3b)i \end{aligned}$$

Since the argument is $-\frac{\pi}{4}$, the real part is positive and the imaginary part is the negative of the real part. Thus

$$2a - 3b = -(-3a - 2b)$$
$$2a - 3b = 3a + 2b$$
$$a = -5b$$

Substituting into the product gives

$$z_1 z_2 = 13b - 13bi$$

Its real part is positive, so $b > 0$. Using the given modulus:

$$\sqrt{(13b)^2 + (-13b)^2} = 26\sqrt{2}$$
$$13\sqrt{2}\,b = 26\sqrt{2}$$
$$b = 2$$
$$a = -5(2) = -10$$

So the answer in the form $a + bi$ is

$$z_2 = -10 + 2i$$

6. The complex number w is given by

$$w = -6\sqrt{3} - 6i$$

(a) (i) Find $|w|$. **[2]**

(ii) Find the principal argument of w, taking $-\pi < \arg(w) \leq \pi$. **[2]**

The complex numbers z_1 and z_2 are given by

$$z_1 = a - 3i, \qquad z_2 = 2(\cos\theta + i\sin\theta)$$

where a is a positive real constant and $-\pi < \theta \leq \pi$.

(b) Given that $z_1 z_2 = w$,

(i) find the exact value of a. **[3]**

(ii) find θ, giving your answer as an exact multiple of π. **[3]**

Solution

(a) (i) Using the real and imaginary parts of w:

$$|w| = 6\sqrt{(\sqrt{3})^2 + 1^2} = 12$$

(ii) w lies in the third quadrant, as its real and imaginary parts are both negative, so

$$\begin{aligned}\arg(w) &= -\pi + \arctan\left(\frac{6}{6\sqrt{3}}\right) \\ &= -\pi + \frac{\pi}{6} \\ &= -\frac{5\pi}{6}\end{aligned}$$

(b) (i) Expanding the product gives

$$z_1 z_2 = 2\big[(a\cos\theta + 3\sin\theta) + (a\sin\theta - 3\cos\theta)i\big]$$

Equating real and imaginary parts with w, then dividing by 2:

$$a\cos\theta + 3\sin\theta = -3\sqrt{3}, \qquad a\sin\theta - 3\cos\theta = -3$$

Squaring and adding eliminates the cross terms:

$$\begin{aligned}(a\cos\theta + 3\sin\theta)^2 + (a\sin\theta - 3\cos\theta)^2 &= 27 + 9 \\ (a^2 + 9)(\cos^2\theta + \sin^2\theta) &= 36 \\ a^2 + 9 &= 36 \\ a^2 &= 27\end{aligned}$$

Since a is positive, taking the positive root gives

$$a = 3\sqrt{3}$$

(ii) Using $a = 3\sqrt{3}$ from part (b)(i), divide by z_1 and multiply numerator and denominator by its conjugate:

$$\begin{aligned} z_2 = \frac{w}{z_1} &= \frac{-6\sqrt{3} - 6i}{3\sqrt{3} - 3i} \\ &= \frac{(-6\sqrt{3} - 6i)(3\sqrt{3} + 3i)}{(3\sqrt{3})^2 + 3^2} \\ &= -1 - \sqrt{3}\,i \end{aligned}$$

Since $z_2 = 2(\cos\theta + i\sin\theta)$, θ is the argument of z_2. z_2 lies in the third quadrant, so

$$\begin{aligned} \theta &= -\pi + \arctan(\sqrt{3}) \\ &= -\pi + \frac{\pi}{3} \\ &= -\frac{2\pi}{3} \end{aligned}$$

7. The complex numbers z and w are given by $z = 3 + i$ and $w = 2 + 4i$.

(a) (i) Find the modulus and principal argument of zw.
(ii) Express zw in the form $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $-\pi < \theta \leq \pi$. **[4]**

(b) The complex number v satisfies

$$\frac{1}{v} = \frac{1}{w} - \frac{1}{z}$$

Find v in the form $a + bi$, where a and b are real. **[5]**

(c) The complex conjugate of v is denoted by v^*.
Show that $vv^* = k$, where k is a real number whose value is to be determined. **[2]**

Solution

(a) (i) Multiplying and using $i^2 = -1$:

$$\begin{aligned} zw &= (3 + i)(2 + 4i) \\ &= 6 + 12i + 2i + 4i^2 \\ &= 2 + 14i \end{aligned}$$

Hence the modulus is

$$|zw| = \sqrt{2^2 + 14^2} = \sqrt{200} = 10\sqrt{2}$$

zw lies in the first quadrant, as its real and imaginary parts are both positive, so its principal argument is

$$\arg(zw) = \arctan\left(\frac{14}{2}\right) = \arctan 7 \approx 1.43 \text{ radians}$$

(ii) Using part (a)(i), $r = 10\sqrt{2}$ and $\theta = \arctan 7$, which lies in the required interval:

$$zw = 10\sqrt{2}(\cos(\arctan 7) + i\sin(\arctan 7))$$

(b) Combining the fractions:

$$\begin{aligned} \frac{1}{v} &= \frac{1}{w} - \frac{1}{z} \\ &= \frac{z - w}{zw} \\ &= \frac{(3 + i) - (2 + 4i)}{2 + 14i} \\ &= \frac{1 - 3i}{2 + 14i} \end{aligned}$$

Taking reciprocals and multiplying numerator and denominator by the conjugate of $1 - 3i$:

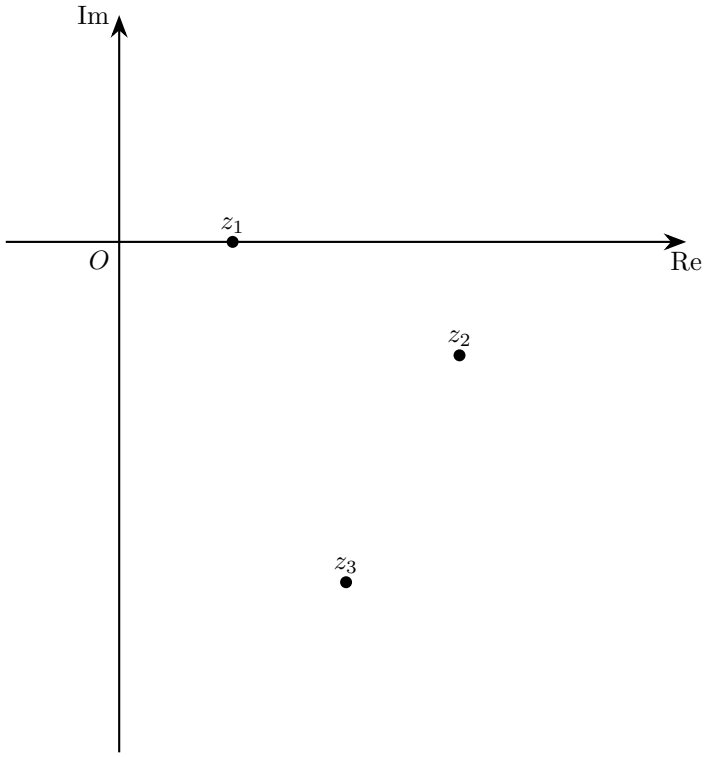
$$\begin{aligned} v &= \frac{2 + 14i}{1 - 3i} \\ &= \frac{(2 + 14i)(1 + 3i)}{(1 - 3i)(1 + 3i)} \\ &= \frac{2 + 6i + 14i + 42i^2}{1 - 9i^2} \\ &= \frac{-40 + 20i}{10} \\ &= -4 + 2i \end{aligned}$$

This is in the form $a + bi$ with $a = -4$, $b = 2$.

(c) The conjugate is $v^* = -4 - 2i$, so

$$\begin{aligned} vv^* &= (-4 + 2i)(-4 - 2i) \\ &= 16 - (2i)^2 \\ &= 16 + 4 = 20 \end{aligned}$$

Thus $vv^* = k$, where $k = 20 \in \mathbb{R}$.



8. The Argand diagram above shows the complex numbers $z_1 = 1$, $z_2 = 3 - i$ and $z_3 = 2 - 3i$.

(a) Explain why no polynomial equation of degree four with real coefficients can have all three of these numbers as roots. **[2]**

(b) Show that

$$\arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) = \frac{\pi}{4}$$ **[3]**

(c) Hence show that

$$\arctan(3) - \arctan\left(\frac{1}{2}\right) = \frac{\pi}{4}$$ **[2]**

(d) Shade, on a copy of the diagram, the region consisting of all points whose complex coordinates satisfy

$$|z - 1| \leq |z - 2 + 3i|$$ **[2]**

Solution

(a) Non-real roots of a polynomial with real coefficients occur in complex conjugate pairs. Thus $3 + i$ and $2 + 3i$ would also have to be roots.

This gives five distinct roots, 1, $3 - i$, $3 + i$, $2 - 3i$ and $2 + 3i$, whereas a polynomial of degree four has at most four distinct roots.

(b) Substituting the given complex numbers and multiplying by the conjugate of the denominator:

$$\begin{aligned}\frac{z_2 - z_1}{z_3 - z_1} &= \frac{2 - i}{1 - 3i} \\ &= \frac{(2 - i)(1 + 3i)}{(1 - 3i)(1 + 3i)} \\ &= \frac{2 + 6i - i - 3i^2}{1 + 9} \\ &= \frac{5 + 5i}{10} = \frac{1}{2} + \frac{1}{2}i\end{aligned}$$

$\frac{1}{2} + \frac{1}{2}i$ lies in the first quadrant, so

$$\arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) = \arctan(1) = \frac{\pi}{4}$$

(c) Both $z_2 - z_1 = 2 - i$ and $z_3 - z_1 = 1 - 3i$ lie in the fourth quadrant, so

$$\arg(z_2 - z_1) = -\arctan\left(\frac{1}{2}\right), \qquad \arg(z_3 - z_1) = -\arctan(3)$$

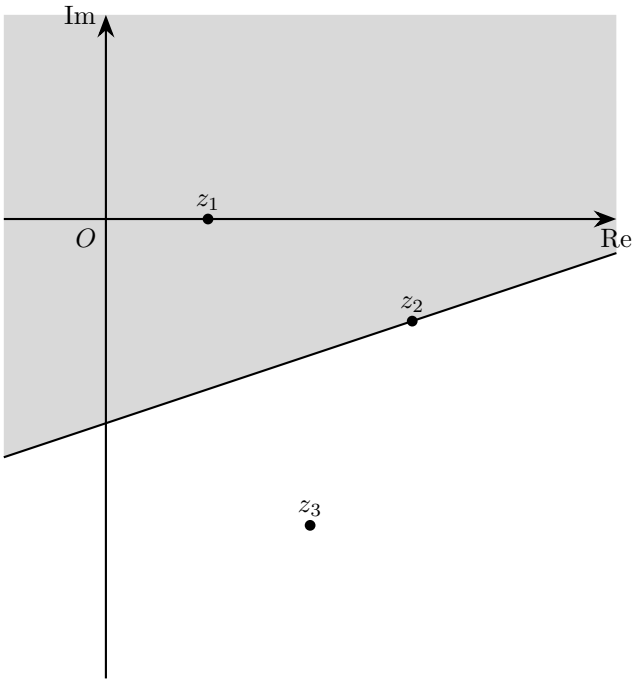
For a quotient, the modulus is the quotient of the moduli and the argument is the difference of the arguments, adjusted by a multiple of 2π to lie in the principal range:

$$\left|\frac{w_1}{w_2}\right| = \frac{|w_1|}{|w_2|}, \qquad \arg\left(\frac{w_1}{w_2}\right) = \arg(w_1) - \arg(w_2)$$

Hence, using part (b):

$$\begin{aligned}\frac{\pi}{4} &= \arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) \\ &= \arg(z_2 - z_1) - \arg(z_3 - z_1) \\ &= \arctan(3) - \arctan\left(\frac{1}{2}\right)\end{aligned}$$

(d) The shaded diagram should look like:



The solid boundary is included. Writing $z = x + iy$, and squaring both non-negative moduli, gives:

$$\begin{aligned} |z-1| \leq |z-2+3i| &\iff (x-1)^2 + y^2 \leq (x-2)^2 + (y+3)^2 \\ &\iff x^2 - 2x + 1 + y^2 \leq x^2 - 4x + 4 + y^2 + 6y + 9 \\ &\iff 2x - 6y \leq 12 \\ &\iff y \geq \frac{x-6}{3} \end{aligned}$$